How-To Library

Informatica

Secure@Source 4.5 Quick Start Deployment Guide: Embedded Hadoop Cluster

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Abstract

This guide describes the installation process for Secure@Source 4.5 in a Linux environment for an embedded Hadoop cluster.

Supported Versions

- Secure@Source 4.5

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Steps to Deploy Secure@Source

After you procure and setup the hardware according to the *Secure@Source Sizing Guide*, perform the following tasks to install Secure@Source.

1. Verify the prerequisites for the Secure@Source server and Informatica Services node.
2. Verify the prerequisites for the Hadoop server and nodes.
3. Verify port availability.
4. Verify the prerequisites for the repository database.
5. Install Secure@Source.
6. Configure connectivity to the data source.

Note: Some steps refer to sections in the *Informatica Secure@Source Installation and Configuration Guide*.

Step 1. Verify Prerequisites for the Secure@Source Server and Informatica Services Node

Operating System

Log in as the root user and verify the following Operating System requirements:

- ❑ A supported Linux distribution is installed. For details, see the *Secure@Source 4.5 Product Availability Matrix* located on the Informatica Network: https://network.informatica.com/docs/DOC-17804
- ❑ The operating system license is registered.

- ❑ The following packages are installed:

❑cURL ❑JAVA **Note:** Informatica recommends version jdk1.8.0_162 ❑rpm ❑scp ❑ssh	❑tar ❑unzip ❑wget ❑zip

Disk Space and Processors

Verify the following Secure@Source server prerequisites:

- ❑ Disk space for the mount point where you plan to install Secure@Source is at least 500 GB.
- ❑ The appropriate number of processors are available, based on the environment sizing agreement.
- ❑ The appropriate number of cores are available, based on the environment sizing agreement.
- ❑ The appropriate amount of memory is available, based on the environment sizing agreement.
- ❑ The appropriate disk space is available, based on the environment sizing agreement.
- ❑ The `/tmp` folder is at least 16 GB.

Create User

- ❑ Create the Secure@Source user to install and run Secure@Source on the Secure@Source server.
- ❑ Set up passwordless ssh between the Secure@Source user on the Secure@Source server and the Hadoop Gateway user on the Hadoop Cluster node.
  To set up the passwordless ssh, complete the following steps:
  1. Log in to the Secure@Source host using the Secure@Source domain user that you will use to install Secure@Source.
  2. Enter: `ssh-keygen -t rsa`
     Accept the default value for each prompt.
  3. Enter: `cd $HOME/.ssh`
  4. Enter: `ssh-copy-id -i id_rsa.pub <HADOOP_GATEWAY_USER>@<HADOOP_HOST>.informatica.com`
  5. To verify that you set up passwordless ssh access correctly, enter the following command: `ssh <HADOOP_GATEWAY_USER>@<HADOOP_HOST>.informatica.com`

Settings

Verify or configure the following settings:

- ❑ The `/etc/hosts` file includes the fully qualified host names of the Secure@Source server, Hadoop server, and all Hadoop nodes.
  Example: `<machine ipaddress> <fully qualified name> <alias>`
- ❑ The LANG environment variable is set to `en_US.UTF-8`.
- ❑ The LC_ALL environment variable is not set.
- ❑ The JRE_HOME environment variable is not set.

- ❑ The following user limits are set:
  - Max user processes: 100000
  - Stack size: 25000
  - Open files: 100000
- ❑ SELinux is set to Permissive on the Secure@Source node.
- ❑ Configure the firewall and verify the necessary ports are open. See "Step 3. Secure@Source and Hadoop Ports" on page 9.
- ❑ Start the ntpd service on the Secure@Source installation node.
- ❑ Verify that the ntpstat utility reports a synchronized status.
- ❑ Disable chronyd, the ntp replacement in RHEL7.x.
- ❑ Set all nodes to the same time zone.

Set Environmental Variables

For Informatica Services	For Oracle	For IBM Db2	For Microsoft SQL Server
❑ JAVA_HOME ❑ LD_LIBRARY_PATH ❑ ODBCHOME ❑ ODBCINI ❑ ORACLE_HOME ❑ PATH	❑ ORACLE_HOME ❑ TNS_ADMIN= $ORACLE_HOME/network/admin/tnsnames.ora ❑ PATH must contain $ORACLE_HOME/bin ❑ LD_LIBRARY_PATH must contain $ORACLE_HOME/lib	❑ DB2_HOME ❑ PATH must contain $DB2_HOME/bin ❑ LD_LIBRARY_PATH must contain $DB2_HOME/lib	❑ ODBCHOME ❑ PATH must contain $ODBCHOME/bin ❑ LD_LIBRARY_PATH must contain $ODBCHOME/lib ❑ ODBCINST must be set to $ODBCHOME/odbcinst.ini ❑ ODBCINI must be set to $ODBCHOME/odbc.ini

Install and Configure the Database Client for the Repository Database

Oracle Settings

If the Secure@Source repository database is Oracle, complete the following steps on the Secure@Source server:

- ❑ Install and configure the Oracle Administrator Client software on the Secure@Source node.
- ❑ Verify that the `tnsnames.ora` file is readable by other users.
- ❑ Verify that the `tnsnames.ora` file has an entry for the repository database.

Sample `tnsnames.ora` file:

```
<addressname> =
(DESCRIPTION =
   (ADDRESS_LIST =
     (ADDRESS = (PROTOCOL = TCP)(Host = <hostname>)(Port = <port>))
   )
(CONNECT_DATA =
   (SERVICE_NAME = <service_name>)
)
)
```

Example:

```
QA12C =
(DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = inkrh7orcl12c.informatica.com)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = QA12C.informatica.com)
    )
  )
```

IBM Db2 Settings

If the Secure@Source repository database is IBM Db2, complete the following steps on the Secure@Source server:

- ❑ Install and configure the IBM Db2 Client software on the Secure@Source node.
- ❑ Verify that the IBM Db2 catalog entry is added for the repository database.
- ❑ Verify that the **NCHAR_MAPPING** variable is set to `GRAPHIC_CU16`.

Step 2. Verify Prerequisites for the Hadoop Server and Nodes

Operating System

Verify the following Operating System prerequisites:

- ❑ A supported Linux distribution is installed. For details, see the *Secure@Source 4.5 Product Availability Matrix* located on the Informatica Network: https://network.informatica.com/docs/DOC-17804
- ❑ Bash is the default shell.
- ❑ Access to a RedHat/SUSE repository.
  If access to public Redhat/SUSE repositories cannot be provided, follow the steps outlined in the following guides to create a local repository:
  - Creating a Yum Repository Server for Red Hat and CentOS
- ❑ The following packages are installed on each Hadoop node:

❑cURL ❑JAVA **Note:** Install Oracle JDK version jdk1.8.0_162. You cannot use OpenJDK. ❑keytool ❑kernel-devel ❑kernel-headers ❑libtirpc-devel ❑lsof ❑OpenSSL ❑For RHEL v6.x: Python, Python-Devl, and Python-Libs v2.6.x ❑For RHEL v7.x: Python, Python-Devel, and Python-Libs v2.7.5-34 or later within the 2.7.x version. **Note:** Ensure that the system does not upgrade the Python packages on the Hadoop node during installation.	❑rpm ❑rsync ❑scp ❑ssh ❑tar ❑unzip ❑wget ❑yum ❑zip ❑If the server is not connected to the internet or the user repository does not have all the packages, verify that postgresql, postgresql-libs, and postgresql-server are installed.

If the operating system is SUSE Linux Enterprise Server, the following additional packages are installed on each Hadoop node:

- ❑ netcat-openbsd
- ❑ php_Curl
- ❑ Zypper

Disk Space

Verify the following Hadoop server prerequisites:

- ❑ The appropriate number of processors are available, based on the environment sizing agreement.
- ❑ The appropriate amount of memory is available, based on the environment sizing agreement.
- ❑ The following directories have the required minimum available disk space:

Directory	Minimum Available Disk Space	Notes
/	60 GB	The Hadoop Service installer requires at least 60 GB on /. However, if you specify a custom directory for HDFS, YARN, and ZOOKEEPER logs and data directories during installation, the installer requires at least 10 GB on / and 50 GB on the custom directory.
/hadoop	400 GB	Refer to the *Secure@Source Sizing Guide*.
/var/log	30 GB	Required for the Hadoop log files.
/opt	10 GB	
/var/lib	10 GB	Required for Hadoop binaries.
/usr	20 GB	
/tmp	8 GB	

Create User

- ❑ Log in to the Hadoop server as the root user.
- ❑ Create the Hadoop Gateway user on the Hadoop Gateway node and on all Hadoop nodes. The user must be a non-root user and must have sudo permissions. Complete the following steps to create the Hadoop Gateway user:
  1. Create a user. Use the following command: `# useradd <username>`
     Example: `# useradd ihsuser`
  2. Specify a password. Use the following command: `# passwd <password>`
     Example: `# passwd ihsuser`
  3. Run visudo to edit the `/etc/sudoers` file.

4. Find the following lines in the file that grant sudo access to users in the group wheel when enabled.

```
## Allows people in group wheel to run all commands
# %wheel        ALL=(ALL)       ALL
```

5. Remove the comment character # at the start of the line: `# %wheel ALL=(ALL) ALL`
6. Add the user you created to the wheel group using the following command: `# usermod -aG wheel <username>`
   Example: `# usermod -aG wheel ihsuser`
7. Test that the updated configuration allows the user that you created to run commands using sudo. Use the following su command to switch to the new user account that you created: `# su ihsuser -`

❑ Set up passwordless ssh access from and to the following users:

- From the Hadoop gateway user on the Hadoop Cluster Gateway host to the gateway user on the Hadoop Cluster node when both the Hadoop Cluster Gateway host and the Hadoop Cluster node are on the same machine.
- From the Hadoop gateway user on the Hadoop Cluster Gateway host to the gateway user on the Hadoop Cluster nodes when both the Hadoop Cluster Gateway host and the Hadoop Cluster nodes are on different machines.

To setup passwordless ssh access, complete the following steps:

1. Log in to the Hadoop host as the Hadoop Gateway user.
2. Enter: `ssh-keygen -t rsa`
   Accept the default value for each prompt.
3. Enter: `cd $HOME/.ssh`
4. Enter: `ssh-copy-id -i id_rsa.pub <HADOOP_GATEWAY_USER>@<HADOOP_HOST>.informatica.com`
5. To verify that you set up passwordless ssh access correctly, enter the following command: `ssh <HADOOP_GATEWAY_USER>@<HADOOP_HOST>.informatica.com`

User Permissions

Verify the following user permissions:

❑ The `/tmp` directory has 777 permissions.

❑ The `/tmp` directory mount options must be set to `exec`.
Use the following command: `sudo mount -o remount,exec /tmp`

❑ The `/var` directory does not have 777 permissions.

Kerberos Requirements

For Secure@Source Service to start in Kerberos mode, create the Secure@Source Elasticsearch directory for every node in the cluster.

1. On the Hadoop box, create the following directory structure:

   ```
   /var/lib/hadoop-yarn/<Secure@Source Service name>_elasticsearch
   ```

   Run the following command on the Hadoop box: :

   ```
   $ sudo mkdir /var/lib/hadoop_yarn/<Secure@Source Service name>_elasticsearch
   $ sudo chown <service cluster name>:1000 /var/lib/hadoop_yarn/sats_elasticsearch
   ```

   Note: The Secure@Source Service name in the commands must be in lower case even if the name contains upper case characters. For example, if the Secure@Source Service name is `SatS`, enter the name as `sats` in the command.
2. Assign ownership of the directory to the `<service_cluster_name>` user.

Verify Settings

Verify or configure the following settings:

- ❑ SELinux is set to Disabled on the Hadoop nodes.
- ❑ Configure the firewall and verify the necessary ports are open. See "Step 3. Secure@Source and Hadoop Ports" on page 9.
- ❑ The JAVA_HOME environment variable is set in .bashrc of the Gateway user.
- ❑ The Gateway user has password-less sudo access.
- ❑ Set umask to 0022 for the root and Gateway user on all Hadoop nodes. If necessary, add `umask 0022` to .bashrc.
- ❑ Set up password-less SSH access from the Gateway user on the Hadoop Gateway node to the Gateway user on all Hadoop nodes.
- ❑ Set the following user limits:
  - Max user processes: 100000
  - Stack size: 25000
  - Open files: 100000
- ❑ The file system that is mounted on the `/tmp` directory has the EXEC flag set.
- ❑ The `/etc/hosts` file has an entry for loopback address.
  Example: `127.0.0.1 localhost localhost.localdomain`
- ❑ The `Defaults requiretty` line is commented out in the `sudoers` file.
- ❑ On Red Hat Enterprise Distribution 7.x, the `Defaults secure_path` line is commented out in the `sudoers` file.
  Example: `#Defaults secure_path = /sbin:/bin:/usr/sbin:/usr/bin`
- ❑ The value in the `Defaults secure_path` line is appended to the PATH environment variable in the `.bashrc` file.
  Example: `PATH=$PATH:$HOME/.local/bin:$HOME/bin:/sbin:/bin:/usr/sbin:/usr/bin`
- ❑ The host name is fully qualified.
- ❑ The `/etc/hosts` file contains the fully qualified domain names of the Hadoop nodes and Secure@Source server.
  Example: `<machine ipaddress> <fully qualified name> <alias>`
- ❑ The `/etc/sysconfig/network` contains the fully qualified host name.
- ❑ Start the ntpd service as the root or Gateway user.
- ❑ Disable chronyd, the ntp replacement in Red Hat Enterprise Distribution 7.x.
- ❑ The network time is synchronized with ntpstat.
- ❑ Set all the nodes to the same time zone.
- ❑ On SUSE Linux Enterprise Server operating systems, the libsnappy package is not installed.

Internet Access

To install rpm packages, the server must have internet access.

Step 3. Secure@Source and Hadoop Ports

Verify that the following TCP and UDP ports are open:

Interface Source	Interface Target	TCP Ports	UDP Ports
Laptop/Workstation, Hadoop Host	App Host {Secure at Source}	6005 through 6009, 6014-6114, 6643, 8443, 9443, 9553, 18295	
Hadoop Host	App Host {Secure at Source}	6013, 6113, 9085, 9090,15380, 15381, 15385, 15386, 18105, 18095, 52652	
App Host {Secure at Source}, Laptop/ Workstation	Hadoop Host	8042, 8044, 8080, 8088, 8090, 19888	
App Host {Secure at Source}	Hadoop Host	2181, 5432, 6188, 6443, 6605, 8010, 8020, 8025, 8030, 8032, 8141, 8188, 8190, 8245, 8440, 8441, 8480, 8481, 8670, 9553, 10020, 19888, 45454, 50010, 50070, 50075, 50090, 50470, 50475, 59925, 60200	59804
App Host {Secure at Source}, Hadoop Host	Repository database server	For example: 1521 for Oracle, 1433 for Microsoft SQL Server	

Step 4. Verify Prerequisites for the Repository Database

Create Users

Create users for the following components:

Component	Tablespace Name
Domain Repository	INFA_DOMAIN
Model Repository	INFA_MRS
Model Repository Monitoring Statistics Service	INFA_MRSMS
Content Management	INFA_CMS
Profiling Warehouse	INFA_DIS
Secure@Source	INFA_SATS
Persistent Data Masking*	INFA_PDM
*Required if you plan to enable Persistent Data Masking.	

IBM Db2 Repository Database Requirements

Setup the national character mapping configuration parameter, NCHAR_MAPPING, to GRAPHIC_CU16.

You can use the following commands:

```
db2 connect to <Secure@Source database name>
db2 update db cfg using NCHAR_MAPPING GRAPHIC_CU16
db2 connect reset
```

Microsoft SQL Server Repository Database Requirements

Complete the following tasks if the Secure@Source repository database is Microsoft SQL Server:

Database Settings

Important: The database collation type must be case sensitive. For example: `COLLATE SQL_Latin1_General_CP1_CS_AS`

Configure the following settings:

- ❑ READ_COMMITTED_SNAPSHOT: ON
  This minimizes locking contention.
- ❑ ALLOW_SNAPSHOT_ISOLATION: ON
- ❑ AUTO_UPDATE_STATISTICS_ASYNC: ON
- ❑ AUTO_UPDATE_STATISTICS: ON

Tablespace Sizes

Assign the following tablespace sizes:

Component	Tablespace Name	Size
Domain Repository	INFA_DOMAIN	8 GB
Model Repository	INFA_MRS	75 GB
Model Repository Monitoring Statistics Service	INFA_MRSMS	25 GB
Content Management	INFA_CMS	25 GB
Data Integration Service and Profiling Warehouse	INFA_DIS	250 GB
Secure@Source	INFA_SATS	50 GB
Persistent Data Masking*	INFA_PDM	25 GB
*Required if you plan to enable Persistent Data Masking.		

User Role

Each user must have the server role `public`.

User Privileges

Each user must have the following privileges:

- ❑ ALTER
- ❑ CONNECT
- ❑ CREATE FUNCTION
- ❑ CREATE SCHEMA
- ❑ CREATE TABLE
- ❑ CREATE VIEW
- ❑ DELETE
- ❑ EXECUTE
- ❑ INSERT
- ❑ REFERENCES
- ❑ SELECT
- ❑ UPDATE

Oracle Repository Database Requirements

Complete the following tasks if the Secure@Source repository database is Oracle:

Database Settings

Configure the following settings:

- ❑ open_cursors parameter: 4000
- ❑ Sessions and processes parameters: 1500
- ❑ SGA: 4000 MB or higher

Tablespace Sizes

Assign the following tablespace sizes:

Component	Tablespace Name	Size
Domain Repository	INFA_DOMAIN	8 GB
Model Repository	INFA_MRS	75 GB
Model Repository Monitoring Statistics Service	INFA_MRSMS	25 GB
Content Management	INFA_CMS	25 GB
Data Integration Service and Profiling Warehouse	INFA_DIS	250 GB
Secure@Source	INFA_SATS	50 GB
Persistent Data Masking*	INFA_PDM	25 GB
*Required if you plan to enable Persistent Data Masking.		

User Privileges

Each user must have the following privileges:

- ❑ CREATE PROCEDURE
- ❑ CREATE SESSION
- ❑ CREATE SEQUENCE
- ❑ CREATE SYNONYM
- ❑ CREATE TABLE
- ❑ CREATE TRIGGER
- ❑ CREATE VIEW

User Resource Limits

Assign the following resource limits:

- ❑ SESSIONS_PER_USER: 100 or higher
- ❑ CPU_PER_SESSION: unlimited
- ❑ CONNECT_TIME: unlimited
- ❑ IDLE_TIME: unlimited

Step 5. Install Secure@Source

Log in as the Informatica user and perform the following tasks to install Secure@Source:

- ❑ Download the Secure@Source software and copy it to the Secure@Source and Data Integration Service servers.
- ❑ Download the Informatica license file and copy it to the same location as the installation software.
- ❑ Download the following packages to the `<Secure@Source_Installer_Location>/Server/LDM/source` directory:
  - `ambaribinaries.tar.gz`
  - `ScannerBinaries.zip`

  Note: Ensure that you download binaries for the correct operating system and version.
- ❑ Run the Secure@Source installer.

For details on running the Secure@Source installer, see the *Informatica Secure@Source Installation Guide*.

Step 6. Configure Connectivity to the Data Source

Connect to the data source that you want to scan. You might need to install or configure additional components to connect to the data source. Then enable the ports to the data source on the Secure@Source and all Hadoop nodes. For example, enable port 1521 for Oracle sources and port 1433 for Microsoft SQL Server sources.

Category	Data Store	Connectivity Requirements
Application Sources	SAP	- Import SAP transports. - Install SAP JCo.
Big Data Sources	Cloudera Navigator	None.

Category	Data Store	Connectivity Requirements
	Hadoop Distributed File System	Create a cluster configuration object in the Administrator Tool.
	Hive	Create a cluster configuration object in the Administrator Tool.
Cloud Sources	Amazon Redshift	- Download the Amazon Redshift JDBC Driver. - Obtain the JDBC URL. - Configure a JDBC Connection With Apache Maven.
	Amazon S3	- Enable account with Amazon Web Services. - Note the Access Key and Secret Key.
	Microsoft Azure SQL Database	Note the Azure Blob Storage name and key.
	Microsoft Azure SQL Data Warehouse	Copy the jar file `mssql-jdbc-6.2.0.jre8.jar` to the following directory: `INFA_HOME/java/lib/ext`
	Salesforce	- Verify Salesforce user account and privileges. - Register the Salesforce SSL certificates with the Informatica domain.
Data Integration	Informatica Big Data Management	- Copy the Informatica Big Data Management jar files to the following directory: `<INFA_HOME>/services/SecureAtSourceService/ROOT/WEB-INF/eclipse/plugins` - Create a Secure@Source temporary folder in the following directory: `<INFA_HOME>/tomcat/temp/` - Restart the Secure@Source Services.
	Informatica Cloud	Import the certificate for Informatica Cloud into the Secure@Source trust store.
	Informatica PowerCenter on IBM Db2 Database	Get the sitekey file.
	Informatica PowerCenter on Microsoft SQL Server Database	Get the sitekey file.
	Informatica PowerCenter on Oracle Database	Get the sitekey file.
	Microsoft SQL Server Integration Services	For repository based service: - Configure the 10.2 supported MITI Agent URL to run on the Microsoft SQL Server Integration Services server. - Assign administrator privileges to the MITI Agent. For file based service: - Configure the 10.2 supported MITI Agent URL to run on any server, client, or accessed box. - Copy the Microsoft SQL Server Integration Services mapping files from Visual Studio to any directory on the Secure@Source domain box.

Category	Data Store	Connectivity Requirements
Database Management Sources	IBM Db2	- Install the database client software. - Configure native connectivity to the database.
	IBM Db2 i5	Install the PowerExchange client.
	IBM Db2 zOS	Install the PowerExchange client.
	Microsoft SQL Server	- Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
	MySQL	- Install the database client software or install the JDBC and ODBC drivers. - Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
	Netezza	- Install the database client software or install the JDBC and ODBC drivers. - Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
	Oracle	- Install the database client software. - Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
	Sybase	- Install the database client software or install the JDBC driver. - Configure ODBC or JDBC connectivity to the database. - Configure permissions and privileges for the database user account.
	Teradata	- Install the database client software or install the JDBC and ODBC drivers. - Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
File Management Sources	File System	Create a mount point to the file system on the Hadoop and Informatica domain nodes.

For details on how to connect to each data source, see the *Informatica Secure@Source Administrator Guide*.

Author